HR EMPLOYEE ATTRITION ANALYSIS

Capstone Project Report

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Project Name: Capstone1_HR_Attrition_Cerlin

1. PROBLEM STATEMENT & BUSINESS OBJECTIVES

Organization facing 16.1% attrition (237 out of 1470 employees). High recruitment cost, loss of knowledge, low productivity. Problem is reactive HR, no data-driven system to identify at-risk employees.

Business Questions:

1. What is overall attrition rate and demographic pattern?

2. Which factors (Income, Overtime, Work-Life Balance) most strongly predict attrition?

3. Can we build ML model to predict risk?

4. What retention strategies for HR?

Objective: Build end-to-end solution using Python, SQL, Power BI.

2. DATASET DESCRIPTION

Source: IBM HR Analytics - WA_Fn-UseC_-HR-Employee-Attrition.csv (Kaggle)

Raw: 1470 rows x 35 columns | Cleaned: 1470 rows x 28 columns

Target: Attrition Yes/No

Attributes: Demographics (Age, Gender, MaritalStatus, DistanceFromHome), Job Related (Department, JobRole, MonthlyIncome, OverTime, YearsAtCompany), Satisfaction (JobSatisfaction 1-4, WorkLifeBalance 1-4), Performance.

Raw File: data/raw/WA_Fn-UseC_-HR-Employee-Attrition.csv

Cleaned File: HR_Attrition_Cleaned.csv

3. DATA CLEANING & PREPROCESSING - BEFORE/AFTER

BEFORE:

1. **EmployeeCount:** All 1 (redundant)
2. **Over18:** All Y (redundant)
3. **StandardHours:** All 80 (redundant)
4. **EmployeeNumber:** ID, not needed
5. No nulls but inconsistent categorical formatting

AFTER (in http://Proof.ipynb):

1. Dropped: EmployeeCount, Over18, StandardHours, EmployeeNumber

2. Attrition Yes/No -> 1/0

3. Outliers in MonthlyIncome, TotalWorkingYears handled via IQR

4. One-Hot Encoding: Department, JobRole, EducationField, Gender

5. Label Encoding: JobSatisfaction, WorkLifeBalance

6. New Feature: IncomePerYear, IsHighOvertimeRisk

7. Train-Test Split 80-20 Stratified

8. Scaling: StandardScaler Screenshot of before/after head included in http://Proof.ipynb

4. EXPLORATORY DATA ANALYSIS (EDA) - KEY FINDINGS

- **Overall Attrition:** 16.12%
- **Department:** Sales highest 20.6%
- **OverTime:** Yes 30.4% vs No 10.4% - 3x higher risk!
- **Income:** < $3000 = 28% attrition
- **Age:** 18-30 group = 28% attrition highest
- **JobRole:** Sales Representative & Lab Technician highest
- **DistanceFromHome:** >20km = 25% attrition
- WorkLifeBalance 1 (Bad) = 31.5% vs Rating 4 = 8.2%

Visuals: Pie chart, Bar plots (Attrition vs OverTime, Dept), Correlation heatmap, Boxplot Income vs Attrition. All in http://Proof.ipynb & Power BI.

5. MACHINE LEARNING - PREDICTIVE MODELING

Model: Random Forest Classifier - n_estimators=100, random_state=42

Reason: Handles categorical+numeric, gives feature importance, robust to overfit.

Performance: Accuracy 86.7%, Precision 72% (Attrition Yes), Recall 54%, F1 61%, ROC-AUC 0.81

Feature Importance Top 6:

1. OverTime_Yes 0.18 - Most Important

2. MonthlyIncome 0.14

3. Age 0.11

4. DistanceFromHome 0.09

5. JobRole_Sales Rep 0.08

6. TotalWorkingYears 0.07

Model can be used monthly to score current employees risk.

6. SQL ANALYSIS & POWER BI DASHBOARD

SQL Schema with CREATE TABLE (Mam pointed missing):

CREATE TABLE hr_attrition (Age INT, Attrition VARCHAR(10), Department VARCHAR(50), JobRole VARCHAR(50), MonthlyIncome INT, OverTime VARCHAR(10), JobSatisfaction INT, WorkLifeBalance INT, DistanceFromHome INT);

4 Queries:

Q1: Attrition Rate by Department

Q2: Avg Monthly Income Attrited vs Retained

Q3: OverTime Impact with percentage

Q4: JobRole wise attrition + satisfaction

Results matched Python.

Power BI Dashboard 5 Pages

Page1 Overview KPI: Total 1470, Attrition 16.1%, Avg Income $6502

Page2 Demographics: Age, Gender, MaritalStatus

Page3 Job Analysis: Dept & JobRole wise attrition (fixed axis labels, added data labels)

Page4 Risk Factors: OverTime vs Attrition, Income vs Attrition, Distance impact

Page5 Satisfaction: JobSatisfaction, WorkLifeBalance, EnvironmentSatisfaction
Added DAX Attrition % measure, slicers for Dept & Gender, tooltips.

7. BUSINESS RECOMMENDATIONS & HR RETENTION STRATEGY

1. OverTime Management (Highest Impact): Audit >5hrs/week for 3+ weeks -> manager review. Reduce attrition 30%. Saves $15000 per employee.

2. Compensation Review: Revise salary bands for < $3000 & Sales Rep, quarterly bonus. Addresses 28% low-income attrition.

3. Targeted Retention: Age 18-30, Distance >20km, Sales Dept - monthly 1-1, mentorship for <2 years, WFH for long commute.

4. WorkLifeBalance=1: Workload redistribution, wellness programs.

5. Predictive System: Deploy Random Forest monthly, HR gets top 10% at-risk list for proactive action, retrain quarterly.

ROI: Reduce 16.1% to 12% = retain 60/year = $900,000 annual saving.

Limitations: Dataset 2015, needs current company validation.

8. CONCLUSION & FUTURE WORK

Conclusion: Successfully built end-to-end HR analytics pipeline Raw -> Cleaning -> EDA -> ML -> SQL -> Power BI -> Recommendations. Proved attrition NOT random, driven by controllable factors OverTime, Income, Work-Life Balance. Model 86.7% accuracy as early warning system.

GitHub Structure:

- http://README.md (overview, files, tools, insights)
- Proof_of_Work.ipynb (all code with outputs)
- data/raw/WA_Fn-UseC_-HR-Employee-Attrition.csv
- data/processed/HR_Attrition_Cleaned.csv
- report/Capstone1_Report_Cerlin.pdf
- dashboard/HR_Attrition_Dashboard.pbix
- presentation/HR_Attrition_Presentation.pptx
- sql/hr_analysis.sql with schema + 4 queries

Future: NLP on feedback, real-time Power BI alerts, A/B testing retention programs, expand with manager ratings.